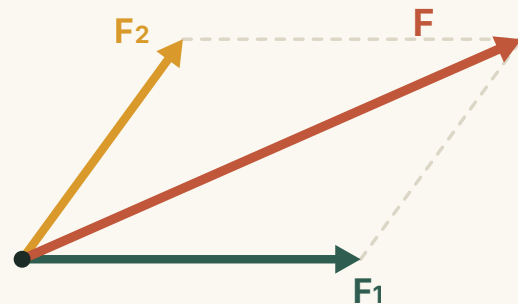


Composing and Resolving Forces

Two ropes can do the work of one. The trick is knowing how the pulls add — and they don't add like ordinary numbers.

ESSENTIAL QUESTION

When can a 3 N pull and a 4 N pull combine to give exactly 5 N — and never 7 N?



Four moves with force arrows.

01

REPLACE

One resultant can stand in for several forces — same effect.

02

COMBINE

At an angle, forces add by the parallelogram rule, not arithmetic.

03

RESOLVE

Run it backwards: split one force into two useful pieces.

04

VECTORS

Why all of this works — force is a vector, not a number.

KEEP THIS IN MIND

Composition and resolution are the **same** rule run in opposite directions.

SET-UP

One force can replace many.

If a single pull would move a body exactly as several pulls do, we call it their **resultant**; the originals are its **components**.

1 Same effect

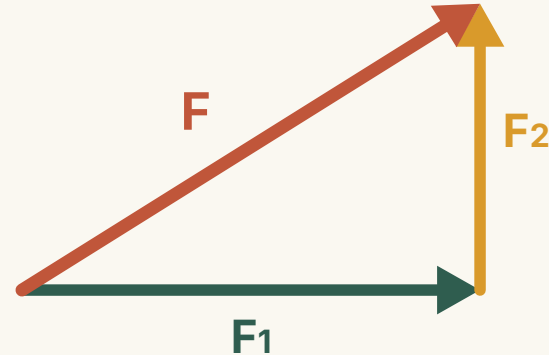
The resultant produces exactly the motion the components together would.

2 Same body

Only forces acting on the one object may be combined this way.

3 Either / or

Use the resultant or the components — never both. It is a replacement, not an extra force.



THE EASY CASE

On one line, just add or subtract.

When two forces point along the same straight line, ordinary arithmetic works — direction decides the sign.

SAME DIRECTION

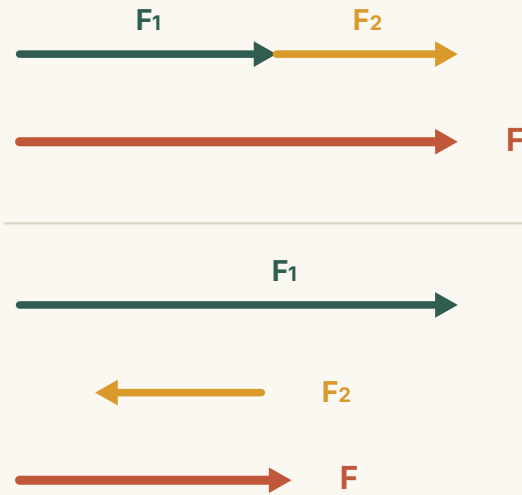
$$F = F_1 + F_2$$

pointing the same way

OPPOSITE DIRECTIONS

$$F = |F_1 - F_2|$$

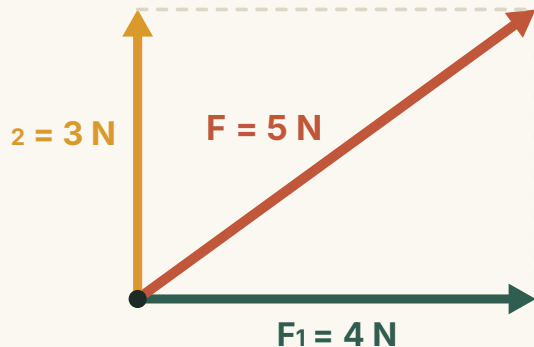
pointing with the larger



THE BIG IDEA

At an angle, the sum shrinks.

Pull a crate 4 N east and 3 N north and you do not get 7 N. The arrows must be joined head-to-tail.



Force has a **direction** as well as a size. Two forces partly pull across each other, so some of the effort is wasted sideways.

SO IN GENERAL

$$F / F_1 \neq F_2$$

THE RULE

The parallelogram rule.

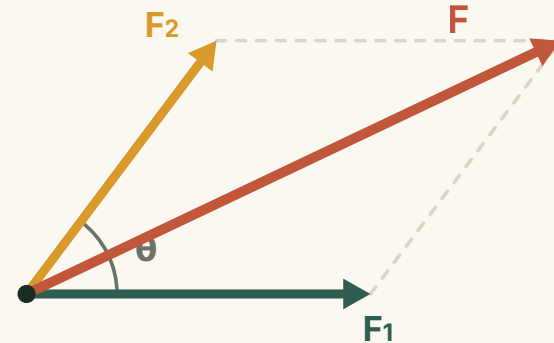
Draw the two forces from one point, complete the parallelogram, and the diagonal is the resultant.

θ the angle between the two forces

F the diagonal of the parallelogram

TRIANGLE SHORTCUT

Join them head-to-tail instead: the closing side is the same resultant.



WHAT THE RULE PROMISES

The resultant lives in a range.

Hold the two sizes fixed and swing the angle: the diagonal can only grow or shrink between two limits.

0° · Largest

Aligned, the resultant is the full sum, $F = F_1 + F_2$.

90° · Right angle

Perpendicular, use Pythagoras: $F = \sqrt{F_1^2 + F_2^2}$.

180° · Smallest

Opposed, it shrinks to the difference, $F = |F_1 - F_2|$.

↕ · In between

A wider angle always gives a smaller resultant.

THE WHOLE STORY IN ONE LINE

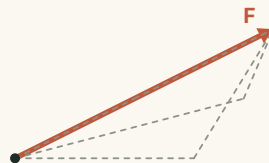
$$|F_1 - F_2| \leq F \leq F_1 + F_2$$

Resolving needs a rule to pin it down.

Splitting one force into two is the reverse of composition — but on its own it has infinitely many answers.

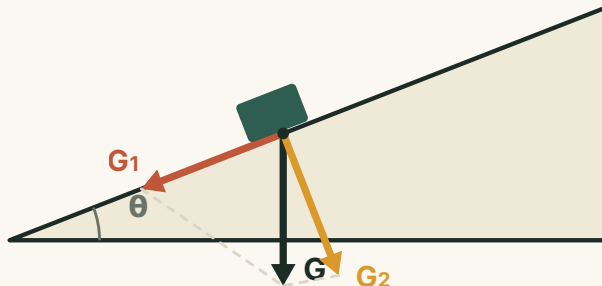
- 1 By effect**
Split along the two directions the force actually acts in.
- 2 By axes**
Split along chosen x and y directions — usually perpendicular.

A single arrow is the diagonal of endless parallelograms.



A block on a slope shows the way.

Gravity does two jobs at once: it drags the block down the slope and presses it into the surface. Split it along those two effects.



DOWN THE SLOPE

$$G_1 = G \sin \theta$$

INTO THE SURFACE

$$G_2 = G \cos \theta$$

Steeper slope $\rightarrow$ bigger $\sin \theta \rightarrow$ the block is pulled down harder.

WORKED EXAMPLE

Three and four make five.

A crate is pulled by 3 N north and 4 N east at right angles. Find the single equivalent pull.

- 1 Right angle, so use Pythagoras**

$$F = \sqrt{F_1^2 + F_2^2} = \sqrt{4^2 + 3^2}$$

- 2 The magnitude**

$F = \sqrt{25} = 5$ N — less than the 7 N you might expect.

- 3 The direction**

$$\tan \alpha = \frac{3}{4} \Rightarrow \alpha \approx 37^\circ \text{ north of east.}$$

THE TAKE-AWAY

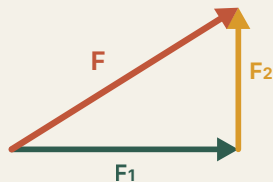
Forces combine like the sides of a **triangle**, not like a shopping bill. Geometry, not arithmetic, sets the total.

- SUMMARY

Force is a **vector**, so it combines by the **parallelogram rule**.

THE ONE FORMULA BEHIND IT ALL

$$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$



The resultant always lands between $|F_1 - F_2| \leq F \leq F_1 + F_2$ — never more, never less.

Resolving is the same rule reversed: split a force where it does its work.